

Machine-Learning Surrogate Optimization of a VPSA Process

SEEN5320 Final Project: CO₂ capture from dry flue gas

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SEEN5320 Final Project

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Goal

Use successful gPROMS VPSA simulations to train a fast surrogate and optimize process KPIs.

- Source data: `data/raw_data.mat`.
- Successful simulations: 1708 rows from an original 2000-point LHS design.
- Inputs: FAD, FRP, FVU, TRP, TED, and TPR.
- Outputs: recovery, purity, productivity, and specific energy.
- Method: tune multiple ML models, select the best Pareto-aware surrogate, then run NSGA-II.

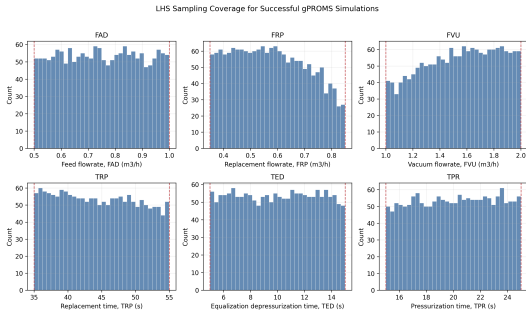
- Wang et al. studied a 3-bed 7-step VPSA process for CO₂/N₂ dry flue gas capture.
- Their ANN surrogate enabled multi-objective optimization without repeatedly solving the full PDAE model.
- This project reproduces that idea with the available scalar simulation data.

Placeholder for literature process schematic and ANN workflow

Data Conversion

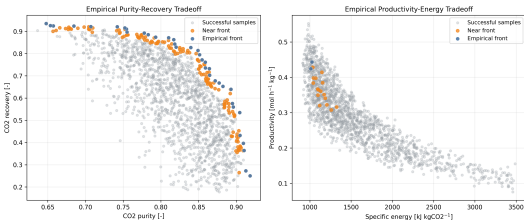
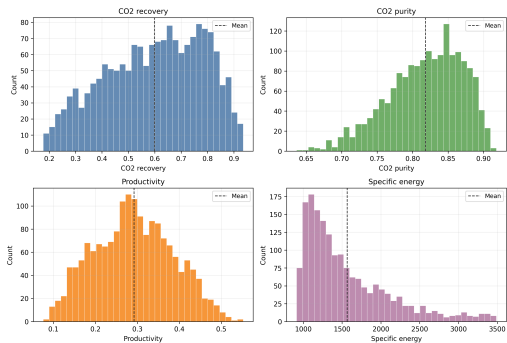
Item	Value
<code>invar</code> shape	1708×6
<code>result_matrix</code> shape	1708×284
KPI columns used	first 4
CSV rows	1708
CSV columns	12

- Failed simulations are absent from the MAT file.
- Remaining profile outputs are ignored in this scalar surrogate version.



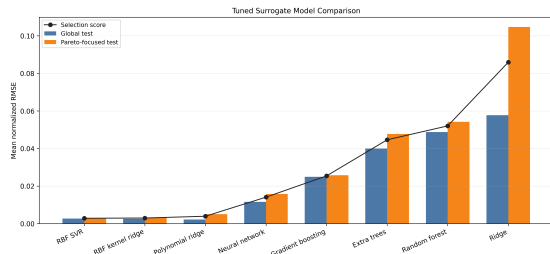
KPI Landscape

Distribution of Four Scalar PSA Performance Indicators



Surrogate Model Benchmark

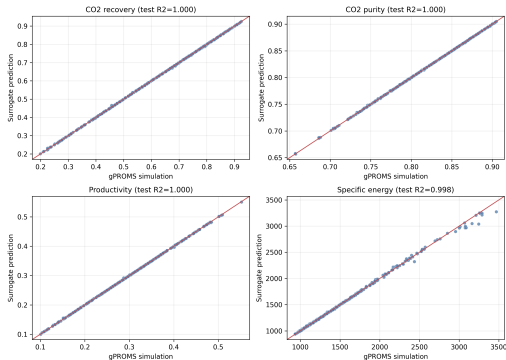
- Same fixed 80/20 train-test split.
- 3-fold CV hyperparameter search on training data.
- Energy modeled in log space.
- Selection score = 0.6 Pareto NRMSE + 0.4 global NRMSE.



Model	Score	R^2
RBF SVR	0.00288	0.99978
RBF KRR	0.00301	0.99977
Poly ridge	0.00395	0.99983
MLP	0.01411	0.99725

Best Surrogate

Best Tuned Surrogate: Held-Out Predictions



Selected model

RBf SVR with $C = 100$, $\epsilon = 0.001$, and $\gamma = 0.1$.

Target	RMSE	Test R^2
Recovery	0.00122	0.99996
Purity	0.00035	0.99995
Productivity	0.00064	0.99995
Energy	22.74	0.99826

Problem 1

$\max(\text{purity}, \text{recovery})$

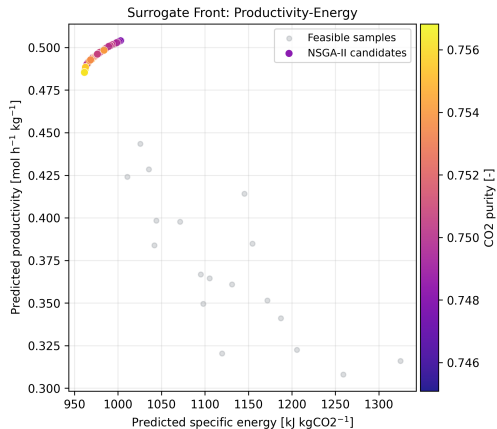
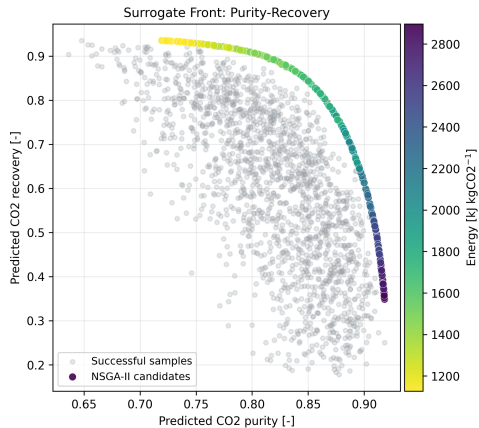
Problem 2

$\max \text{productivity}, \quad \min \text{energy}$

$\text{purity} \geq 0.70, \quad \text{recovery} \geq 0.90$

- Population size: 160.
- Generations: 250.
- Operators: simulated binary crossover and polynomial mutation.
- Ranking: constrained nondominated sorting plus crowding distance.

Optimization Fronts



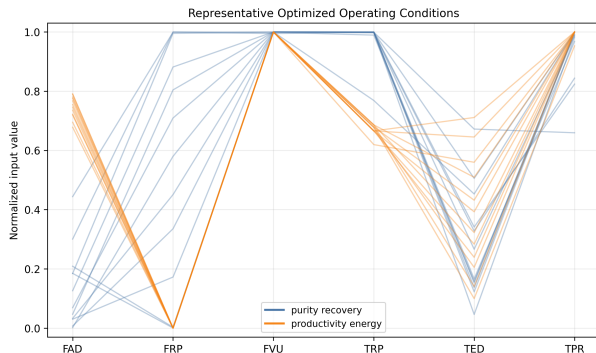
Purity-recovery front

- Purity range: 0.719–0.918.
- Recovery range: 0.349–0.935.
- Maximum recovery point: purity 0.719, recovery 0.935, energy 1139 kJ kgCO₂⁻¹.

Productivity-energy front

- All nondominated candidates satisfy the constraints.
- Productivity range: 0.485–0.504 mol h⁻¹ kg⁻¹.
- Energy range: 961–1003 kJ kgCO₂⁻¹.

Representative Conditions



- Purity-recovery candidates span a broad operating envelope.
- Productivity-energy candidates concentrate near the recovery constraint.
- Optimized points are screening decisions and should be validated by detailed gPROMS reruns.

- The MAT file was converted into a clean scalar table with 1708 successful VPSA simulations.
- Eight surrogate families were tuned and compared with global and Pareto-focused metrics.
- RBF SVR was selected as the best optimization surrogate.
- NSGA-II produced interpretable Pareto fronts for both objective formulations.
- The workflow now connects data conversion, surrogate learning, optimization, figures, report, and slides.